DES Pune University, Pune
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Subject: FCP		
Assignment No.:	2	
Title:	**To develop a C++ program to add constructors and destructor for initializing and destroying objects.**	
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Contents

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Theory(Logic)

This C++ program shows the use of constructors and destructors within a class. We first create a class named Test, which contains a constructor, a destructor, and a function named loud(). A integer variable named Count is declared globally. This variable keeps track of how many objects of the Test class are currently alive. Every time an object is created, the constructor is called automatically, and it increments the Count variable and prints the current number of objects. When an object is destroyed, which happens automatically when it goes out of scope, the destructor is called, printing the number of objects before destruction and then decrementing the count. In the main() function, six objects of the class Test are declared: t, t1, t2, t3, t4, and t5. As soon as each object is created, the constructor runs, so six messages are printed indicating the increasing number of objects. After all objects are created, the function loud() is called on the second object t1, which prints the message "just trying". When the program reaches the end of the main() function, all six objects go out of scope and are destroyed in reverse order of their creation. As a result, six destructor messages are printed, each showing the number of remaining objects before the destruction.

Pseudo Code

Step 1: Include <iostream>
Step 2: Use the standard namespace
Step 3: Declare static integer Count = 0
Step 4: Create a class named Test
Constructor:
Increment Count
Output "No. of objects: " followed by Count
Destructor:
Output "No. of Object destroyed: " followed by Count
Decrement Count
Function loud ():
Output "just trying"
Step 5: Start main function:
Declare six objects of class Test: t, t1, t2, t3, t4, t5
Call the loud function on t1
Step 6: End main function

Program Code

```
#include <iostream>
using namespace std;

static int Count = 0;

class Test {
public:
   Test() {
      Count++;
      cout << "No. of objects: " << Count << "\n";
   }

   ~Test() {
      cout << "No. of Object destroyed: " << Count << endl;
      Count--;
   }

   void loud() {
      cout << "just trying\n";
   }
};

int main() {
   Test t, t1, t2, t3, t4, t5;
   t1.loud();
}
```

Output

```
No. of objects: 1
No. of objects: 2
No. of objects: 3
No. of objects: 4
No. of objects: 5
No. of objects: 6
just trying
No. of Object destroyed: 6
No. of Object destroyed: 5
No. of Object destroyed: 4
No. of Object destroyed: 3
No. of Object destroyed: 2
No. of Object destroyed: 1
```

Conclusion

Code is run successfully, required output is received.